

Question

In liquid ammonia at 238 K, sodium and white phosphorus react at a certain molar ratio to form a yellow crystal. Further analysis indicates that this yellow crystal is a 1 : 1 ionic compound; both its cation and anion are mononuclear sodium coordination ions, each with a coordination number of 5. In this reaction, white phosphorus is converted solely into the phosphorus-containing ligand in the anion of the ionic compound. The molar mass of this ligand is 96 g/mol, and all phosphorus atoms satisfy the octet rule.

In the chemical equation for the formation of this yellow crystal, what is the sum of the coefficients of all substances on both sides of the equation (using the simplest integer ratio for the coefficients)?

A. 13

B. 15

C. 17

D. 19

E. 21

F. 23

G. 25

H. 27

I. 29

J. 31

K. 33

L. 35

M. 37

N. 39

O. 63

P. 65

Q. 67

R. 69

S. 81

T. 83

U. 85

V. 87

W. 43

X. None of the above

Answer

Correct Answer: **T**

Detailed Explanation

In a system containing only the four elements Na, P, H, N, the phosphorus-containing ligand in the anion should consist solely of P, H, N elements. Based on molar mass, the possible chemical formulas for the ligand are PN_4H_9^- , $\text{P}_2\text{N}_2\text{H}_6$, or $\text{P}_3\text{H}_3^{2-}$. Considering the octet rule for phosphorus, the ligand cannot be PN_4H_9^- . If the ligand were $\text{P}_2\text{N}_2\text{H}_6$, the complex formed with sodium ions could not carry a negative charge. Therefore, the chemical formula of the ligand must be $\text{P}_3\text{H}_3^{2-}$.

CHECKPOINT

1 PTS

The ligand's chemical formula is $\text{P}_3\text{H}_3^{2-}$

Since the cation is five-coordinate, the cation is $\text{Na}(\text{NH}_3)_5^+$.

CHECKPOINT

0.5 PTS

The cation is $\text{Na}(\text{NH}_3)_5^+$

Given that the anion is five-coordinate and the $\text{P}_3\text{H}_3^{2-}$ ligand in the anion is chelating, the remaining three coordination sites are occupied by NH_3 , making the anion $\text{Na}(\text{P}_3\text{H}_3)(\text{NH}_3)_3^-$.

CHECKPOINT

1 PTS

The anion is $\text{Na}(\text{P}_3\text{H}_3)(\text{NH}_3)_3^-$

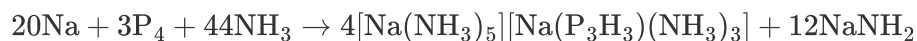
Thus, the chemical formula of the yellow crystal is $[\text{Na}(\text{NH}_3)_5][\text{Na}(\text{P}_3\text{H}_3)(\text{NH}_3)_3]$. The chemical formula of white phosphorus is P_4 .

CHECKPOINT

0.5 PTS

The chemical formula of the yellow crystal is $[\text{Na}(\text{NH}_3)_5][\text{Na}(\text{P}_3\text{H}_3)(\text{NH}_3)_3]$

The left side of the equation consists of Na , P_4 , NH_3 , while the right side includes $[\text{Na}(\text{NH}_3)_5][\text{Na}(\text{P}_3\text{H}_3)(\text{NH}_3)_3]$ and NaNH_2 . Balancing the equation yields:



CHECKPOINT

1 PTS

The equation is $20\text{Na} + 3\text{P}_4 + 44\text{NH}_3 \rightarrow 4[\text{Na}(\text{NH}_3)_5][\text{Na}(\text{P}_3\text{H}_3)(\text{NH}_3)_3] + 12\text{NaNH}_2$

The sum of the coefficients is 83, so option T is correct.

CHECKPOINT

0.5 PTS

The sum of coefficients is 83